

Security configurations

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Configure the router.

a. Console into the router and enable privileged EXEC mode.

router> **enable**

b. Enter configuration mode.

router# **config terminal**

c. Assign a device name to the router.

router(config)# **hostname R1**

d. Set the router's domain name as ccna-lab.com.

R1(config)# **ip domain name ccna-lab.com**

e. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.

R1(config)# **no ip domain lookup**

Encrypt the plaintext passwords.

```
R1(config)# service password-encryption
```

Configure the system to require a minimum 12-character password.

```
R1(config)# security passwords min-length 12
```

Configure the username **SSHadmin** with an encrypted password of **55Hadm!n2020**.

```
R1(config)# username SSHadmin secret 55Hadm!n2020
```

Generate a set of crypto keys with a 1024 bit modulus

```
R1(config)# crypto key generate rsa modulus 1024
```

Assign the privileged EXEC password to **\$cisco!PRIV***

```
R1(config)# enable secret $cisco!PRIV*
```

Assign \$cisco!!CON* as the console password, configure sessions to disconnect after four minutes of inactivity, and enable login.

```
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# exec-timeout 4 0
R1(config-line)# login
```

Assign \$cisco!!VTY* as the vty password, configure the vty lines to accept SSH connections only, configure sessions to disconnect after four minutes of inactivity, and enable login using the local database.

```
R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# exec-timeout 4 0
R1(config-line)# transport input ssh
R1(config-line)# login local
```

Create a banner that warns anyone accessing the device that unauthorized access is prohibited.

```
R1(config)# banner motd $ Authorized Users Only! $
```

Set the clock on the router.

```
R1# clock set 15:20:00 5 Sept 2019
```

Save the running configuration to the startup configuration file.

```
R1# copy running-config startup-config
```

Basic configuration of Adaptive Security Appliance (ASA)

- 1. Bring up the interface – no shut**
- 2. Assign an IP address to the interface of ASA – Ip address <IP add subnetmask>**
- 3. Assign a nameif to the ASA interface – nameif NAME**
- 4. Assign a security level to the interface – Security-level <Value>**

The security level is an integer value ranging from 0 to 100. It tells the trustworthiness of an interface i.e which interface is most trusted. 0 means less trusted while 100 means the most trusted.

If we provide the name INSIDE to an interface, it will automatically provide security level 100 to it and if we provide any other name like OUTSIDE or DMZ, it will assign automatically 0 to it but can be changed manually.

```
asa(config)#int e0
```

```
asa(config-if)#no shut
```

```
asa(config-if)#ip address 192.168.1.1 255.255.255.0
```

```
asa(config-if)#nameif INSIDE
```

```
asa(config-if)#security-level 100
```

Setting passwords –As ASA is a security device, by default it will ask for a password while we try to enter privilege mode. By default, no password is set therefore by simply clicking enter, we can enter the privilege mode.

enable password – The enable password is used for securing privilege mode. In routers, this password is shown in clear text in running configuration but in ASA, this password is encrypted (therefore no enable secret is required.)

```
asa(config)#enable password Cisco      asa(config)#enable password
```

login password –

This password is used for taking access to ASA by using Telnet or SSH. By default, the login password is “Cisco”. We can change it by the command

```
asa(config)#password GeeksforGeeks  or  asa(config)#passwd GeeksforGeeks
```


Using a local database for login:

A local database is configured on the device (username and password) so that it can be used for login purposes. It is configured in the same manner as it is configured on the router. A local database can be configured on the device using the command

```
asa(config)#username BCAS password GeeksforGeeks
```

```
asa(config)#aaa authentication serial console LOCAL
```

Here, note that LOCAL is case-sensitive

ASA is a complex device that requires a good understanding of networking and security concepts to configure properly. This can make it challenging for some users to set up and manage the device.

Lab Activities

User LAN network:

Subnet: 192.168.0.0 /24

Gateway: 192.168.0.1 (ASA inside interface)

LAN-host (for testing): 192.168.0.200

DMZ1 network:

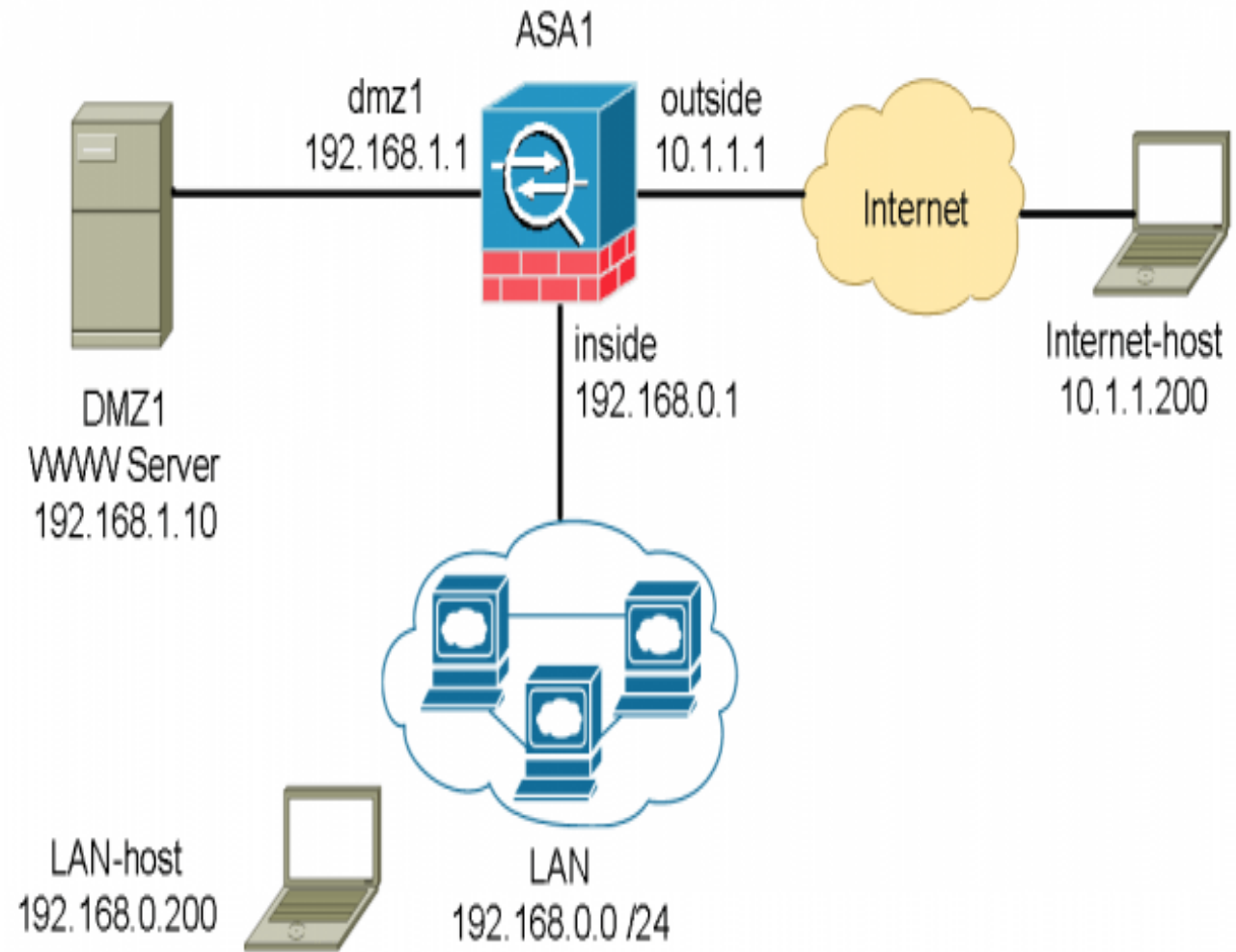
Subnet 192.168.1.0 /24

Gateway: 192.168.1.1

Web server: 192.168.1.10

Internet:

Internet-host (for testing): 10.1.1.200



Thank you